

PRESIDENCY UNIVERSITY, BENGALURU

School of Engineering | Department of CSE (AI & ML)

CA3 Assignment — Literature Survey

Course Code: CAI2500 — Machine Learning

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Topic: Crop Recommendation System Using Machine Learning

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1. Introduction

India is an agrarian economy meaning that its main economy is from agriculture. However the main issue that farmers face is primarily uncertainty of natural occurrences and secondarily crop selection as farmers trust traditional knowledge over data driven reality. Selecting an inappropriate crop for given soil and climatic conditions leads to poor yields, financial losses, and environmental degradation through the misuse of fertilizers and water resources.

Machine learning provides an effective approach to this problem by analyzing patterns in historical agricultural data and suggesting appropriate crops based on soil characteristics and climatic conditions. This literature review examines ten recent studies published between 2020 and 2025 in reputed sources such as IEEE, Springer, and Nature Scientific Reports, focusing on the latest advancements in machine learning-based crop recommendation systems.

2. Research Paper Summaries

Article 1

Citation: *T. O. Mufamadi and R. Ajoodha, "Crop Recommendation Using Machine Learning Algorithms and Soil Attributes Data," in Proc. IEEE IGARSS, 2020, pp. 1–5.*

This paper verifies the usage of various ml classifiers to recommend suitable crops based on multiple soil attributes data collected from diverse agriculture fields in South Africa.

Mufamadi and Ajoodha compared Decision Tree, Naive Bayes, and Random Forest classifiers on chemical properties of soil such as nitrogen, phosphorus, potassium, and pH out of which the Random Forest classifier achieved the highest recommendation accuracy, demonstrating highest efficiency.

Dataset Used: South African soil attribute dataset comprising NPK values and pH readings from multiple agricultural zones

Algorithm / Methodology Used: Decision Tree, Naive Bayes, Random Forest

Performance Metrics / Results: Accuracy, Precision, Recall Score

Limitations of This Work: Dataset is region-specific to South Africa and is not suitable to other climatic and soil conditions

Article 2

Citation: *S. Jain and D. Ramesh, "Machine Learning Convergence for Weather-Based Crop Selection," in Proc. IEEE Int. Students Conf. on Electrical, Electronics and Computer Science (SCECS), 2020, pp. 1–6.*

This paper proposes a weather-driven crop selection system which integrates historical meteorological data with its soil parameters to train classification models. The study uses Logistic Regression, K-Nearest Neighbours, and Support Vector Machine to recommend crops suited to local climate conditions of the Indian agricultural regions. Experimental results show that SVM with an RBF kernel delivers superior performance, and the authors also clearly state that combining both weather and soil data will improve recommendation reliability compared to only soil approach.

Dataset Used: Indian agricultural dataset with temperature, rainfall, humidity, and soil NPK readings

Algorithm / Methodology Used: Logistic Regression, KNN, Support Vector

Performance Metrics / Results: Accuracy

SVM achieved approximately 97% accuracy

Limitations of This Work: Weather data used is historical and does not incorporate real-time data

Article 3

Citation: R. Gupta et al., "WB-CPI: Weather Based Crop Prediction in India Using Big Data Analytics," *IEEE Access*, vol. 9, pp. 137869–137885, 2021, doi: 10.1109/ACCESS.2021.3117247.

This paper exhibits a large-scale weather-based crop prediction system for Indian regions using big data analytics. The authors gathered multiple year's meteorological and soil data from multiple Indian states and then applied ensemble learning techniques like Random Forest and Gradient Boosting to recommend crops. The proposed solution also includes feature importance analysis to identify which climatic variables, finding that monsoon rainfall and temperature are the most decisive factors.

Dataset Used: Multi-state Indian agricultural dataset spanning 10

Algorithm / Methodology Used: Random Forest, Gradient Boosting, Hadoop-based distributed processing

Performance Metrics / Results: Accuracy 96.3%, F1-score

Limitations of This Work: High complexity due to high data

Article 4

Citation: N. N. Thilakarathne, M. S. A. Bakar, P. E. Abas, and H. Yassin, "A Cloud Enabled Crop Recommendation Platform for Machine Learning-Driven Precision Farming," *Sensors*, vol. 22, no. 16, p. 6299, 2022, doi: 10.3390/s22166299.

This paper talks about a cloud-integrated crop recommendation platform that's primary aim is to collect the real-time soil and environmental data using IoT sensors and then processes it using machine learning models that are deployed on a cloud backend. The architecture uses Random Forest as the primary classifier as it achieved highest accuracy. The cloud deployment helps in assuring scalability.

Dataset Used: Real-time IoT sensor data including soil moisture, NPK levels, temperature, and humidity.

Algorithm / Methodology Used: Random Forest, SVM and Decision Tree

Performance Metrics / Results: Accuracy 99%

Limitations of This Work: Over Reliance on IoT infrastructure.

Article 5

Citation: S. P. Raja, B. Sawicka, Z. Stamenkovic, and G. Mariammal, "Crop Prediction Based on Characteristics of the Agricultural Environment Using Various Feature Selection Techniques and Classifiers," *IEEE Access*, vol. 10, pp. 23625–23641, 2022, doi: 10.1109/ACCESS.2022.3154350.

This study conducts an analysis on feature selection methods combined with various classifiers that are used for crop prediction. The authors evaluate Filter, Wrapper, and Embedded feature selection methods along with various classifiers such as Random Forest, SVM, and Naive Bayes. Random Forest provides the most accuracy, and that unnecessary or irrelevant features degrade the performance of the classifier.

Dataset Used: Agricultural environment dataset with 22 crop labels; features include N, P, K, temperature, humidity, pH, and rainfall.

Algorithm / Methodology Used: Random Forest, SVM, Naive Bayes

Performance Metrics / Results: Accuracy up to 99.1% with Wrapper feature selection and Random Forest

Limitations of This Work: Wrapper feature selection is expensive for large datasets

Article 6

Citation: M. Y. Shams, S. A. Gamel, and F. M. Talaat, "Enhancing Crop Recommendation Systems with Explainable Artificial Intelligence: A Study on Agricultural Decision-Making," *Neural Comput. Appl.*, vol. 36, no. 11, pp. 5695–5714, 2024, doi: 10.1007/s00521-023-09391-2.

This paper highlights on the interpretable gap in the current crop recommendation models by including the Explainable AI technique alongwith classifiers. The authors apply SHAP (SHapley Additive exPlanations) and LIME to state the predictions generated by ensemble models such as XGBoost and Random Forest. Results confirm that XGBoost achieves over 99% accuracy while SHAP visualizations reveal that rainfall and potassium concentration are the most influential predictors.

Dataset Used: Crop Recommendation Dataset (Kaggle); 2200 records covering 22 crop types with 7 soil and climate features

Algorithm / Methodology Used: XGBoost, Random Forest; SHAP and LIME for explainability

Performance Metrics / Results: Accuracy 99.5%, Precision 99.4%, Recall 99.4%, F1-score 99.4%

Limitations of This Work: SHAP explanations add computational overhead during inference

Article 7

Citation: A. Gosai, C. Raval, R. Nayak, and H. Jayswal, "Crop Recommendation System Using Machine Learning," in *Proc. 10th Int. Conf. on System Modelling and Advancement in Research Trends (SMART)*, 2021, vol. 3307, pp. 554–557.

This paper provides us with a crop recommendation system that targets small and marginal farmers of Gujarat, India. The authors also design a lightweighted pipeline with the help of Decision Tree and Random Forest classifiers which where trained on local soil and climate data. Analysis confirms that Random Forest performs better than the single Decision Tree in all evaluation metrics.the paper also suggest the idea of using a web interface to connect farmers

Dataset Used: Local Gujarat agricultural dataset with NPK, temperature, humidity, pH, and rainfall features; approximately 1500 records

Algorithm / Methodology Used: Decision Tree, Random Forest

Performance Metrics / Results: Random Forest accuracy: 98.2%; Decision Tree accuracy: 90.5%

Limitations of This Work: Dataset is limited to a single Indian state, so it cant be used for all states or regions

Article 8

Citation: P. Naga Srinivasu, M. F. Ijaz, and M. Wozniak, "XAI-Driven Model for Crop Recommender System for Use in Precision Agriculture," *Comput. Intell.*, vol. 40, no. 1, pp. 1–26, 2024, doi: 10.1111/coin.12629.

This paper suggest an XAI-driven crop recommender which combines a fully stacked ensemble classifier with SHAP-based explanation modules to provide transparent precision agriculture recommendations. The proposed stacking architecture is of multiple layers that include multiple base learners including Random Forest, Gradient Boosting which helps in achieving higher accuracy .The authors also show that stacking helps in reducing variance while SHAP values emboses the non-linear interaction between soil pH and rainfall.

Dataset Used: Kaggle Crop Recommendation Dataset supplemented with regional Indian soil survey data; 2200+ records

Algorithm / Methodology Used: Stacking ensemble (Random Forest, Gradient Boosting, KNN as base, Logistic Regression as meta-learner)

Performance Metrics / Results: Accuracy 99.6%, F1-score 99.5%

Limitations of This Work: Stacking increases training and inference time

Article 9

Citation: O. Turgut, I. Kok, and S. Ozdemir, "AgroXAI: Explainable AI-Driven Crop Recommendation System for Agriculture 4.0," in *Proc. IEEE Int. Conf. on Big Data (BigData)*, 2024, doi: 10.1109/BigData62323.2024.10825771.

The paper presents a well structured Agriculture 4.0 framework that involves IoT sensor, cloud computing concepts and explainable AI to deliver real-time crop recommendations. The system employs XGBoost as its primary classifier and uses LIME for generation of local explanations for every recommendation delivered to farmers using a mobile app. The authors alsoorganized multiple user studies with farmers to show that LIME explanations significantly increase trust in the system unlike black box that reduce explainability.

Dataset Used: Multi-source IoT agricultural dataset including soil sensors and weather stations; 1 million+ time-series data points

Algorithm / Methodology Used: XGBoost

Performance Metrics / Results: Accuracy 98.7

Limitations of This Work: Rscaling and application in rural areas are tough

Article 10

Citation: D. Gosai et al., "High-Precision Crop Recommendation System with Stacking Ensemble Classifiers for Optimizing Agricultural Productivity," *Sci. Rep.*, vol. 15, 2025, doi: 10.1038/s41598-025-09640-5.

This paper suggests a proposes a high-precision crop recommendation system that employs feature fusion and stacking ensemble classifiers to optimize agricultural productivity. The authors demonstrate that feature fusion — combining raw soil parameters with derived interaction features — enables all stacked ensemble classifiers to exceed the accuracy of individually trained models while significantly reducing overfitting. Experiments across multiple Indian state datasets confirm that the stacking approach achieves the highest accuracy and precision among all compared techniques, establishing a new performance benchmark for crop recommendation systems.

Dataset Used: Multi-state Indian agricultural dataset covering diverse soil types and climate zones; crop recommendation Kaggle dataset used for benchmarking

Algorithm / Methodology Used: Stacking ensemble classifiers, Random Forest, XGBoost

Performance Metrics / Results: Accuracy exceeding 99.5%, Precision 99% ,Recall 99%

Limitations of This Work: increases preprocessing complexity

4. References

- [1] T. O. Mufamadi and R. Ajoodha, "Crop Recommendation Using Machine Learning Algorithms and Soil Attributes Data," in *Proc. IEEE IGARSS*, 2020, pp. 1–5.
- [2] S. Jain and D. Ramesh, "Machine Learning Convergence for Weather-Based Crop Selection," in *Proc. IEEE Int. Students Conf. on Electrical, Electronics and Computer Science (SCEECS)*, 2020, pp. 1–6.
- [3] R. Gupta et al., "WB-CPI: Weather Based Crop Prediction in India Using Big Data Analytics," *IEEE Access*, vol. 9, pp. 137869–137885, 2021, doi: 10.1109/ACCESS.2021.3117247.
- [4] N. N. Thilakarathne, M. S. A. Bakar, P. E. Abas, and H. Yassin, "A Cloud Enabled Crop Recommendation Platform for Machine Learning-Driven Precision Farming," *Sensors*, vol. 22, no. 16, p. 6299, 2022, doi: 10.3390/s22166299.
- [5] S. P. Raja, B. Sawicka, Z. Stamenkovic, and G. Mariammal, "Crop Prediction Based on Characteristics of the Agricultural Environment Using Various Feature Selection Techniques

and Classifiers," IEEE Access, vol. 10, pp. 23625–23641, 2022, doi: 10.1109/ACCESS.2022.3154350.

[6] M. Y. Shams, S. A. Gamel, and F. M. Talaat, "Enhancing Crop Recommendation Systems with Explainable Artificial Intelligence: A Study on Agricultural Decision-Making," Neural Comput. Appl., vol. 36, no. 11, pp. 5695–5714, 2024, doi: 10.1007/s00521-023-09391-2.

[7] A. Gosai, C. Raval, R. Nayak, and H. Jayswal, "Crop Recommendation System Using Machine Learning," in Proc. 10th Int. Conf. on System Modelling and Advancement in Research Trends (SMART), 2021, vol. 3307, pp. 554–557.

[8] P. Naga Srinivasu, M. F. Ijaz, and M. Wozniak, "XAI-Driven Model for Crop Recommender System for Use in Precision Agriculture," Comput. Intell., vol. 40, no. 1, pp. 1–26, 2024, doi: 10.1111/coin.12629.

[9] O. Turgut, I. Kok, and S. Ozdemir, "AgroXAI: Explainable AI-Driven Crop Recommendation System for Agriculture 4.0," in Proc. IEEE Int. Conf. on Big Data (BigData), 2024, doi: 10.1109/BigData62323.2024.10825771.

[10] D. Gosai et al., "High-Precision Crop Recommendation System with Stacking Ensemble Classifiers for Optimizing Agricultural Productivity," Sci. Rep., vol. 15, 2025, doi: 10.1038/s41598-025-09640-5.

5. Research Gap

Across the reviewed studies, most crop recommendation systems rely heavily on traditional machine learning and ensemble models using soil and weather data, often achieving high accuracy. However, several clear research gaps remain. Many models are trained on region-specific or limited datasets, making them less generalizable across different climates and soil types. Real-time data integration is still limited, with many systems depending on historical datasets instead of live IoT inputs. While recent works introduce explainable AI techniques, most systems are still complex, computationally heavy, and not easily deployable for small-scale farmers. Additionally, there is a lack of fully scalable, user-friendly solutions that combine high accuracy, real-time adaptability, and interpretability in a single framework suitable for practical agricultural use in diverse regions.